Sec: SR STAR - A & APEX, ALL IITS | DT: 06 / 01 /2026
Time: 3 Hrs | GTM – 10 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
A	C	A	A	C	D	D	A	D	B
11	12	13	14	15	16	17	18	19	20
A	C	C	B	A	D	A	D	B	C
21	22	23	24	25					
2	15	6	2	3					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
C	C	B	A	B	D	C	A	C	D
36	37	38	39	40	41	42	43	44	45
B	B	D	C	C	D	C	B	A	C
46	47	48	49	50					
923	40	67	4	6					

MATHEMATICS

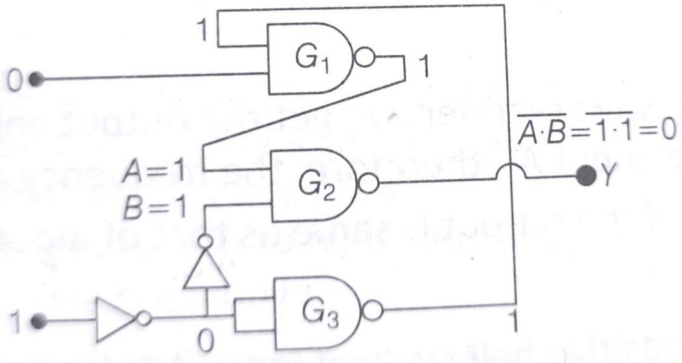
51	52	53	54	55	56	57	58	59	60
C	A	B	B	C	C	A	B	B	A
61	62	63	64	65	66	67	68	69	70
B	C	B	C	B	C	A	B	D	B
71	72	73	74	75					
9	64	18	9	7					

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HINTS

PHYSICS

1. Lower NOT gate inverts input zero, NOT gate from NAND gate inverts this output to 1. Upper NAND gate
Convers this input 1 and input 0 to 1. This A=1 and B=1 become inputs of NAND gate giving final input as zero. Choice A is correct



2. $v_t = \frac{2}{9}\frac{(\sigma-\rho)}{\eta} r^2 g$

$= \frac{2}{9}[2520-1260]\frac{(0.003)^2 \times 10}{1.260}$

$v_t = 2\ cm/s$

$v_t = \frac{s}{t} = \frac{10}{2}$

$v_t = 5\,\text{sec}$

3. $Torque\ \tau = (2F)R + F\left(\frac{R}{2}\right) + FR(-1) = \frac{3FR}{2}$

4. $P = \frac{1}{f}, \frac{1}{f} = (\mu-1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$R_1 = R_1 \quad R_2 = -R$

$P = \frac{1}{f} = (\mu-1)\left(\frac{2}{R}\right) \ldots\ldots\ldots\ldots 1$

$For\ part-A: R_1 = R, \quad R_2 = -R$

$P_A = \frac{1}{f} = (\mu-1)\left(\frac{2}{R}\right)$

$P_A = P$

$For\ part-B: R_1 = R, \quad R_2 = -\alpha$

$P_B = \frac{1}{f} = (\mu-1)\left(\frac{1}{R} - \frac{1}{\alpha}\right)$

$P_B = P/2$

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5. $V = \dfrac{k(p.r)}{r^3} = \dfrac{k(p\hat{k}).(x\hat{i}+y\hat{j}+z\hat{k})}{(x^2+y^2+z^2)^{3/2}}$

$\dfrac{p}{4\pi\varepsilon_0}\dfrac{z}{(x^2+y^2+z^2)^{3/2}}$

6. As there is a hole at minima point P_2, the hole will act as a source of fresh light for the slits S_3 and S_4. Therefore, there will be a regular two slit pattern on the second screen. Hence, option (d) is correct.

7. Internal energy and volumes are state variables while work done by gas and heat absorbed are process variables

8. $F_{avg} = \dfrac{\Delta p}{\Delta t} = \dfrac{mv_0 - (-mv_0)}{\frac{\pi m}{qB}} = \dfrac{2qB_0v_0}{\pi}$

9. The fringe width due to introduction of transparent sheet does not change fringe width, $\omega = \lambda D/d$ and is proportional to wavelength

10. $|\varepsilon_{avg}| = \left|\dfrac{\Delta\phi}{\Delta t}\right| = \left|\dfrac{0-2}{6}\right| = \dfrac{1}{3}V$

12. Using work energy theorem

$W_{mg} + W_T = \Delta K$

$\Rightarrow -mgL(1+\cos\theta) + 0 = \dfrac{1}{2}mv_B^2 - \dfrac{1}{2}m(\sqrt{3gL})^2$

$\Rightarrow v_B^2 = 3gL - 2gL(1+\cos\theta).....(i)$

Equation for centripetal force,

$T + mg\cos\theta = \dfrac{mv_B^2}{L}$

when string becomes slack, T=0

$0 + mg\cos\theta = \dfrac{m}{L}(3gL - 2gL(1+\cos\theta))$

$\Rightarrow \cos\theta = \dfrac{1}{3}$ and $v_B = \sqrt{3gL - 2gL(1+\dfrac{1}{3})} = \dfrac{gL}{3}$

Once string gets slack, the particle will become projectile motion,

$H_{\max} = \dfrac{u^2\sin^2\theta}{2g} = \dfrac{\left(\sqrt{\frac{gL}{3}}\right)^2\left(1-\frac{1}{9}\right)}{2g} = \dfrac{4L}{27}$

Maximum height above A,

$\Rightarrow L + L\cos\theta + H_{\max} = L + \dfrac{L}{3} + \dfrac{4L}{27} = \dfrac{40L}{27}$

13. The effective gravity in the frame of carriage is the sum of g downwards and a upwards resulting in a net upwards direction

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Statement 1 is true

Time period now will be, $T = 2\pi\sqrt{\frac{l}{a-g}}$

Statement 2 is false

14. At resonance, $X_L = X_C \Rightarrow Z = R$

$$i = \frac{V_R}{R} = \frac{100}{1000} = 0.1\ A$$

$$X_L = X_C = \frac{1}{\omega c}$$

$$= \frac{1}{200 \times 2 \times 10^{-6}} = 2500\ \Omega$$

$$V_L = iX_L = 0.1 \times 2500 = 250\ V$$

15. $$F = -\eta A \frac{dV}{dx}$$

$$\frac{F}{A} = -\eta \frac{dV}{dx}$$

$$P = \eta \frac{V}{L}$$

$$\eta = \frac{PL}{V} = PLV^{-1}$$

16. Applying equation of continuity,

$$A_1V_1 = A_2V_2$$

$$\frac{\pi(2)^2 \times 3}{4} = 100 \times \frac{\pi(0.05)^2}{4} v_2$$

$$v_2 = 48\ m/s$$

18. $$\rho = \frac{M}{V} = \frac{M}{\pi r^2 l} = \frac{1}{\pi} M\ \ r^{-2}\ \ l^{-1}$$

$$\Rightarrow \frac{\Delta\rho}{\rho} = \frac{\Delta M}{M} + \frac{2\Delta r}{r} + \frac{\Delta l}{l} = \frac{0.1}{6.7} + \frac{2 \times 0.001}{0.287} + \frac{0.01}{3.28}$$

$$= \frac{0.1}{67} + \frac{2}{287} + \frac{1}{328} = 0.025$$

$$\therefore\ percentage\ error = \frac{\Delta\rho}{\rho} \times 100 = 0.025 \times 100 = 2.5\%$$

19. $$C = C_v + \frac{R}{1-n}$$

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$$C=\frac{R}{\gamma-1}+\frac{R}{1-n}<0$$

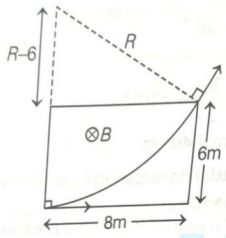
$$\frac{(\gamma-n)R}{(\gamma-1)(1-n)}<0$$

$As,\ \gamma>1, \gamma>n>1$

20. $F_y=2y^3$

$$w=\frac{1}{2}(81-0)=40.5\ J$$

21.

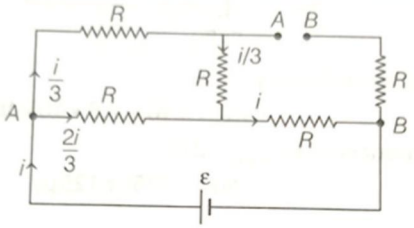


$$(R-6)^2+8^2=R^2 \Rightarrow R=\frac{25}{3}m$$

$$R=\frac{mv}{qB}$$

$$=\frac{25}{3}=\frac{5\times10^{-3}\times v}{12\times10^{-6}\times100}=v=2m/s$$

22.



In steady state, capacitor acts as an open circuit

$$i=\frac{\varepsilon}{R_{eq}}=\frac{\varepsilon}{\left(\frac{5R}{3}\right)}=\frac{3\varepsilon}{5R}$$

$$V_A-V_B=\left(\frac{i}{3}\right)R+iR=\frac{4}{3}iR$$

$$V_A-V_B=\frac{4}{3}\left(\frac{3\varepsilon}{5R}\right)R=12 \text{ b}$$

$$\varepsilon=15V$$

23. $q=CV \Rightarrow \frac{dq}{dt}=C\frac{dV}{dt}=\frac{A\varepsilon_0}{d}\frac{dV}{dt}$

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$$= \frac{3.14\times(2\times10^{-2})^2\times8.85\times10^{-12}}{0.1\times10^{-3}}\times5\times10$$

$$= 5.56\times10^{3}\,A$$

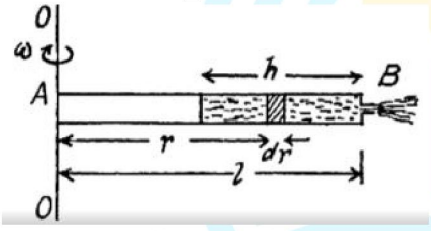
24. $$\frac{d_p}{d_r} = \rho r\omega^2$$

$$d_p = \rho r\omega^2 d_r$$

$$\int_{\rho_0}^{\rho} d_p = \int_{(l-h)}^{r} \rho r\omega^2 d_r$$

$$\rho(r) = \rho_0 + \rho\omega^2[r^2 - (l-h)^2]$$

$$\rho(l) = \rho_0 + \frac{\rho\omega^2}{2}[(l^2 - (l-h)^2]$$



By Bernoulli's equation

$$\rho_0 + \frac{1}{2}\rho\omega^2[2lh - h^2] = \rho_0 + \frac{1}{2}\rho v^2$$

$$v = \omega h\sqrt{\frac{2l}{h} - 1}$$

25. $$F_{CA} = -\left(\frac{\mu_0 i_A i_c l}{2\pi r_{AC}}\right) = -\frac{4\pi\times10^{-7}\times10\times10\times0.15}{2\pi(0.1)}$$

$$= -3\times10^{-5}\,N$$

$$F_{CB} = -\frac{\mu_0 i_B i_c l}{2\pi r_{BC}} = 6\times10^{-5}\,N$$

$$Net\ \ force\ on\ C = F_{CA} + F_{CB} = -3\times10^{-5}\,N(right)$$

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CHEMISTRY

26. Sulphur is converted to Na_2S

27. $H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + H_2O$

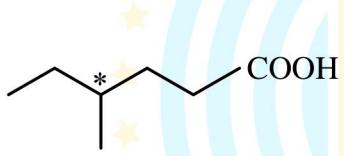
Moles	0.2	0.2	–
After reaction moles	0.1	–	0.1

Molarity of $Na_2SO_4 = \frac{0.1}{4} = 0.025\,M$

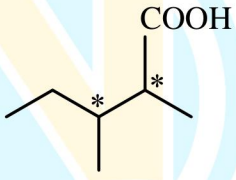
$[H_2SO_4] = \frac{0.1}{4} = 0.025\,M$

Mass of Na_2SO_4 formed $= 0.1 \times 142 = 14.2g$

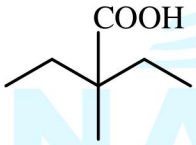
31.



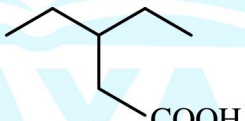
(2)



(4)



(1)



(1)

32. In option (c), F_2 is reduced and oxygen is oxidised.

33. $Pb(OH)_2(s) \rightleftharpoons Pb^{2+}(aq) + 2OH^-(aq)$

s 2s

$K_{sp} = 4s^3$

$K_{sp} = 4(6.7 \times 10^{-6})^3$

$K_{sp} = 1.203 \times 10^{-15}$

$[Pb^{2+}] = \frac{K_{sp}}{[OH^-]^2} = \frac{1.203 \times 10^{-15}}{(10^{-6})^2} = 1.203 \times 10^{-3}\,M$

34. $[Ni(CO)_4]$ has tetrahedral geometry

35. 1) Isothermal process : $\Delta T = 0$ and $\Delta U = 0$

2) $\Delta U = q + W$ and $W = -P\Delta V$, $\Delta V = 0 \Rightarrow \Delta U = q$

3) Adiabatic process $q = 0$

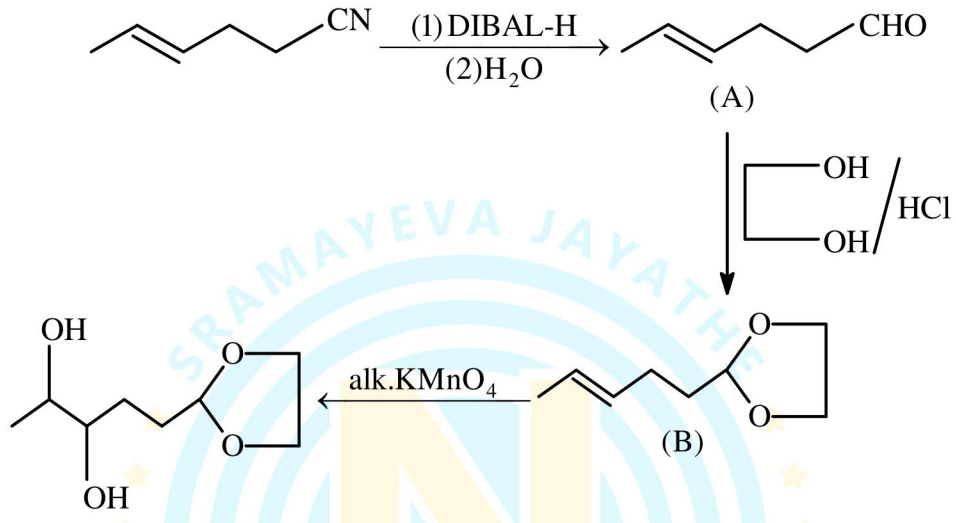
4) Free expansion $P_{ext} = 0$

36. I, III, IV, V and VI are reducing sugars

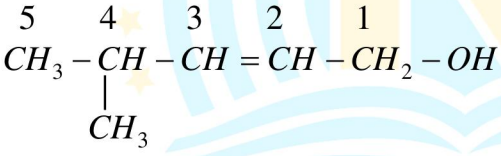
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38. Conceptual

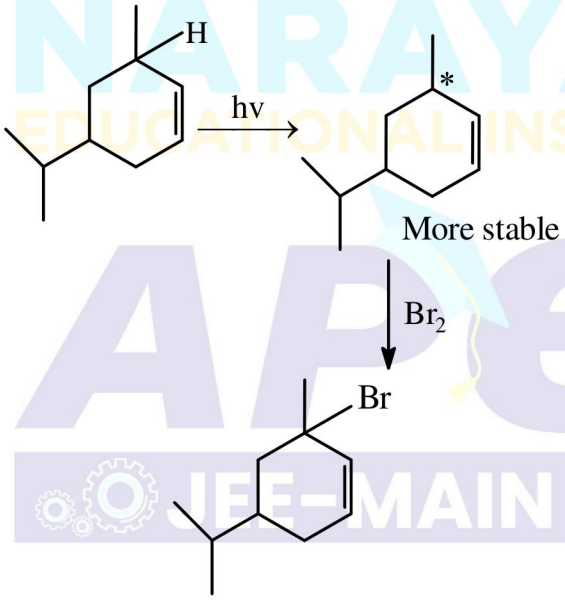
39.



40.

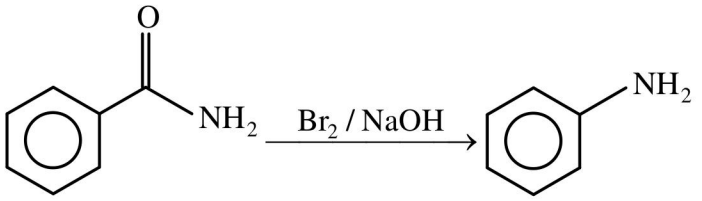


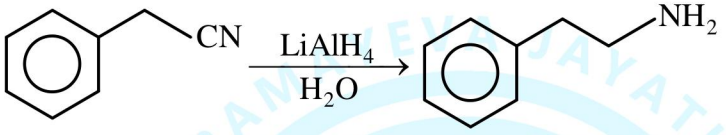
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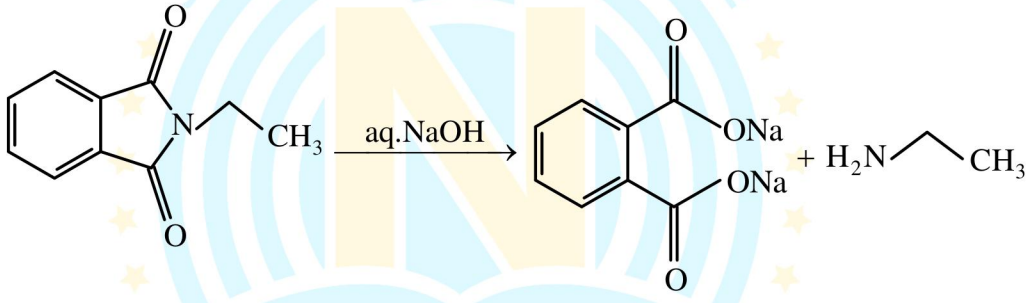


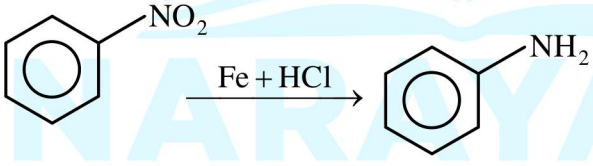
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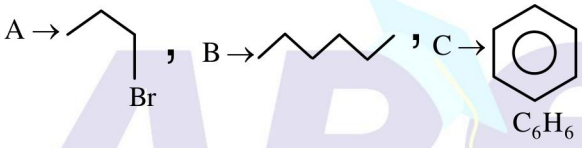








46.



Percentage of carbon $= \frac{72}{78} \times 100 = 92.3$

47. $H_3PO_3 + 2KOH \longrightarrow K_2HPO_3 + 2H_2O$

$$\frac{M_1V_1}{n_1} = \frac{M_2V_2}{n_2}$$

$$\frac{0.1 \times 20}{1} = \frac{0.1 \times V_2}{2}$$

$\therefore\ V_2 = 40\,mL$

48. $2NaCl(s) \longrightarrow 2Na(s) + Cl_2(g)\ ;\quad \Delta H^{\ominus} = 822\,kJ$

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$H_2SO_4(\ell) \longrightarrow H_2(g) + S(s) + 2O_2(g)$; $\Delta H^{\ominus} = +811$ kJ

$2Na(s) + S(s) + 2O_2(g) \longrightarrow Na_2SO_4(s)$; $\Delta H^{\ominus} = -1382$ kJ

$H_2(g) + Cl_2(g) \longrightarrow 2HCl(g)$; $\Delta H^{\ominus} = -184$ kJ

$2NaCl + H_2SO_4 \longrightarrow Na_2SO_4 + 2HCl$; $\Delta H^{\ominus} = 1633 - 1566 = 67$ kJ

49.

50. ZnS, MnS, COS and Nis are formed
x = 4
ZnS and MnS are soluble in conc. HCl
y = 2.

MATHEMATICS

51. $|\text{adj}(\text{adj}(A))| = |A|^{(n-1)^2}$
$\Rightarrow |A| = x + y + z = 12$
$x \geq 1, y \geq 1, z \geq 1$
$\Rightarrow {}^{11}C_2 = 55$.

52. $5! \cdot 3! \cdot 2! = 1440$

53. $(1+x^2)^2(1+x)^n = (1+2x^2+x^4)({}^nC_0 + {}^nC_1x + {}^nC_2x^2 + ...)$
$= {}^nC_0 + {}^nC_1x + ({}^nC_2 + 2 \cdot {}^nC_0)x^2 + ...$
Hence, $A_0 = 1, A_1 = {}^nC_1, A_2 = {}^nC_2 + 2$ which are in A.P.

54. Let a general point on line L is $(2t+2, 3-3t, t-1)$, DRs of AP $2t-, 4-3t, t-5$,
for this point to be point P, as AP $\perp$ Line L

$$2(2t-1) - 3(4-3t) + (t-5) = 0 \Rightarrow t = \frac{19}{14} \Rightarrow P\left(\frac{33}{7}, \frac{15}{14}, \frac{5}{14}\right)$$

55. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
$\frac{5}{8} = \frac{3}{8} + \frac{4}{8} - P(A \cap B)$.
$\Rightarrow P(A \cap B) = \frac{2}{8} = \frac{1}{4}$.
Now, $P\left(\frac{A^c}{B}\right) = \frac{P(A^c \cap B)}{P(B)} = \frac{P(B) - P(A \cap B)}{P(B)} = 1 - \frac{2}{8} \cdot \frac{8}{4} = \frac{1}{2}$.
Now, $2P\left(\frac{A}{B^c}\right) = \frac{2P(A \cap B^c)}{P(B^c)} = \frac{2(P(A) - P(A \cap B))}{1 - P(B)} = 4\left(\frac{3}{8} - \frac{2}{8}\right) = \frac{1}{2} \Rightarrow$ (a) is

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correct

$$P\left(\frac{A}{B}\right)=\frac{P(A\cap B)}{P(B)}=\frac{1}{4}\cdot\frac{2}{1}=\frac{1}{2}=P(B)\Rightarrow\text{(b) is correct}$$

Again $P\left(\frac{A^c}{B^c}\right)=\frac{P(A^c\cap B^c)}{P(B^c)}=\frac{1-P(A\cup B)}{1-P(B)}=2\left(1-\frac{5}{8}\right)=\frac{3}{4}.$

$$P\left(\frac{B}{A^c}\right)=\frac{P(B\cap A^c)}{1-P(A)}=\frac{P(B)-P(A\cap B)}{5/8}=\frac{\frac{1}{2}-\frac{1}{4}}{\frac{5}{8}}=\frac{1}{4}\cdot\frac{8}{5}=\frac{2}{5}.$$

Hence $8P\left(\frac{A^c}{B^c}\right)=15P\left(\frac{B}{A^c}\right)\Rightarrow$ (c) is not correct

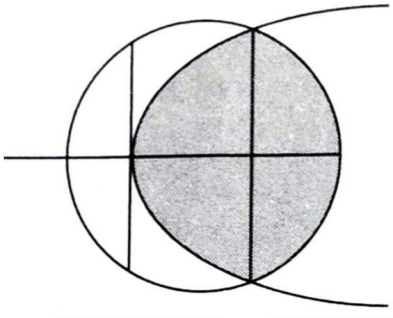
Again $2P\left(\frac{A}{B^c}\right)=\frac{1}{2}$ [From eq(1)] $\Rightarrow P\left(\frac{A}{B^c}\right)=\frac{1}{4}=P(A\cap B)\Rightarrow$ (d) is correct

56. Note that $\sec^{-1}\sqrt{1+x^2}=\tan^{-1}x$; $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)=2\tan^{-1}x$ for $x>0$.

$$I=\int\frac{e^{\tan^{-1}x}}{1+x^2}((\tan^{-1}x)^2+2\tan^{-1}x)dx \text{ put } \tan^{-1}x=t.$$

$$=\int e^t(t^2+2t)dt=e^t\cdot t^2=e^{\tan^{-1}x}(\tan^{-1}x)^2+C$$

57. Circle intersect the $y^2=4x$ at $(1,\pm2)$ end of latus rectum



Required curve $=2\int_0^1 2\sqrt{x}+2\pi=\frac{8}{3}+2\pi$.

58. $f'(x)=0$ and apply wavy curve method

59. Let $P\equiv(at_1^2,2at_1)$ and $Q\equiv(at_2^2,2at_2)\Rightarrow t_1t_2=-1$

Equation of AP is $y=\frac{2}{t_1}x_1\Rightarrow R\equiv(-a,-2a/t_1)$.

Similarly, $T\equiv\left(-a,\frac{-2a}{t_2}\right)$

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$$\text{Slope of RS} = \frac{\left(\frac{-2a}{t_1}\right)}{-2a} = \frac{1}{t_2}.$$

$$\text{Slope of TS} = \frac{\left(\frac{-2a}{t_2}\right)}{-2a} = \frac{1}{t_2}$$

$$\text{Now, } \left(\frac{1}{t_1}\right)\times\left(\frac{1}{t_2}\right) = \frac{1}{t_1 t_2} = -1$$

$$\Rightarrow \angle RST = 90^0.$$

60. $|\vec{a}+\vec{b}|^2 < 1$

$|\vec{a}|^2 + |\vec{b}|^2 + 2\vec{a} - \vec{b} < 1$

$1+1+2\vec{a}-\vec{b} < 1$

$\vec{a}-\vec{b} < -\frac{1}{2}$

61. $\lim\limits_{x\to 0}\left(e^{\frac{2}{x}\ln(1+x)-2}\right)^{\frac{4}{\sin x}} = e^{\lim\limits_{x\to 0}\frac{4}{\sin x}\left(e^{2\left(\frac{\ln(1+x)}{x}-1\right)}-1\right)}.$

$$= e^{\lim\limits_{x\to 0}\frac{4}{\sin x}\left(\frac{e^{2\left[\frac{\ln(1+x)}{x}-1\right]}-1}{2\left(\frac{\ln(1+x)}{x}-1\right)}\times 2\left[\frac{\ln(1+x)}{x}-1\right]\right)}.$$

$$= e^{\lim\limits_{x\to 0} 8\left(\frac{x-\frac{x^2}{2}+\ldots.}{x}-1\right)\times\frac{1}{\sin x}}$$

$$= e^{-\frac{8}{2}} = e^{-4}.$$

62. $f(1)\Rightarrow$ odd
$f(2)\Rightarrow$ even
$f(3)\Rightarrow$ odd
$f(4)\Rightarrow$ even
} Hence, 4 functions.

$f(1)\Rightarrow$ even
$f(2)\Rightarrow$ odd
$f(3)\Rightarrow$ even
$f(4)\Rightarrow$ odd
} Hence, 4 functions

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$f(1) \Rightarrow$ odd
$f(2) \Rightarrow$ even
$f(3) \Rightarrow$ even
$f(4) \Rightarrow$ odd
} Hence, 4 functions

63. $t_1 + t_3 + t_5 + ... + t_{2n+1} = \frac{n+1}{2}[2a + n(2d)] = 248$

$t_2 + t_4 + t_6 + ... + t_{2n} = \frac{n}{2}[2(a+d) + (n-1)2d] = 217$

$t_{2n+1} - t_1 = 2n \cdot d = 56$

$\Rightarrow \frac{n+1}{2}[2a + 56] = 248$ and $\frac{n}{2}[2a + 56] = 217$

$\Rightarrow n = 7, a = 3.$

64. $\lim_{t \to x} \frac{t^3 f(x) - x^3 f(t)}{t^2 - x^2} = \frac{1}{2} \Rightarrow 3x f(x) - x^2 f'(x) = 1$

$\Rightarrow \frac{dy}{dx} - \frac{3y}{x} = \frac{-1}{x^2} \Rightarrow y = \frac{1}{4x} + \frac{3}{4}x^2 \quad (\because f(1) = 1)$

65. $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{vmatrix} = 0$

66. $x^3 dy + 3x^2 y dx = y^2 dx + 2xy\, dy$

$d(x^3 y) = d(xy^2)$

$\int d(x^3 y) = \int d(xy^2) \Rightarrow x^3 y = xy^2.$

67. Minimum occurs at the vertex

Vertex $\in [0, 2]$

Maximum occurs at one of end points $x = 0$ or $x = 2$

Vertex $x = -\frac{a}{2}$

Minimum value $= b - \frac{a^2}{4} = 2$

$a \le -\frac{a}{2} \le 2 \Rightarrow -4 \le a \le 0$

Maximum value at end points

$f(0) = b,\ f(2) = 4 + 2a + b$

Maximum value three, so one of these values each to 3

Option (a) $(-2, 3)$

$f(0) = 3 \Rightarrow f(2) = 3$

Vertex at $x = 1$

$f(1) = 2$

Satisfies all conditions

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68. $(\log_{\sqrt{3}} \tan x)\left(\sqrt{\frac{\log_{\sqrt{3}} 3}{\log_{\sqrt{3}} \tan x}+\frac{\log(\sqrt{3})^3}{\log \sqrt{3}}}\right)=-1$.

$(\log_{\sqrt{3}} \tan x)\left(\sqrt{\frac{2}{\log_{\sqrt{3}} \tan x}+3}\right)$

Let $\log_{\sqrt{3}} \tan x = y \Rightarrow y\sqrt{\frac{2}{y}+3}=-1 \Rightarrow y=-1 \ (\because \ y<0)$

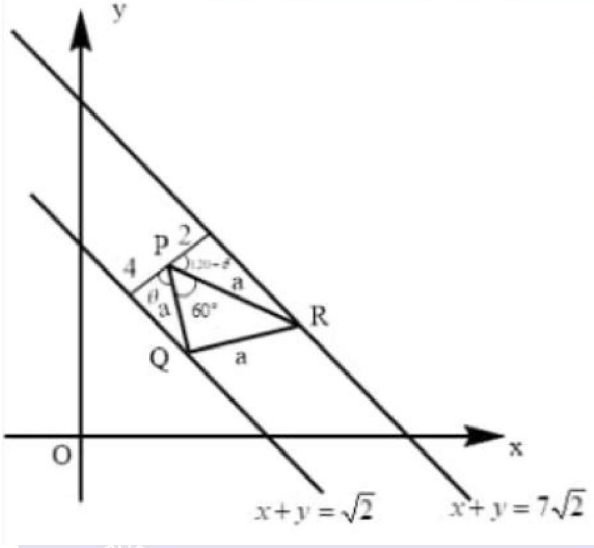
$\Rightarrow \tan x=\frac{1}{\sqrt{3}} \Rightarrow x=\frac{\pi}{6},\frac{7\pi}{6}$.

69. Clearly $z=-\sqrt{3}-3i \Rightarrow \frac{z}{2\sqrt{3}}=-\left(\frac{1}{2}+\frac{i\sqrt{3}}{2}\right)$.

70. $\cos\theta=\frac{4}{a}$

$\cos(120-\theta)=\frac{2}{a}$

Solve them $a=\frac{4\sqrt{7}}{\sqrt{3}}$



71. $2be=2Be^1=2\sqrt{5}$

$be=Be^1=\sqrt{5}$

$\Rightarrow \frac{e}{e^1}=\frac{B}{b}$

$\Rightarrow \frac{1}{2}=\frac{B}{b}$

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$\Rightarrow b = 2B$

$b+ = B = 6 \Rightarrow 2B + B = 6 \Rightarrow 3B = 6 \Rightarrow B = 2$

$b = 6 - B = 6 - 2 = 4$

$(be)^2 = b^2 - a^2 \Rightarrow 5 = 16 - a^2 \rightarrow a^2 = 16 - 5 = 11$

$(Be^1)^2 = B^2 + A^2 \Rightarrow 5 = 4 + A^2 \Rightarrow A^2 = 1$

$$\left|LR - LR^1\right| = \left|\frac{2a^2}{b} - \frac{2A^2}{B}\right| = \left|\frac{2(11)}{4} - \frac{2(1)}{2}\right| = \left|\frac{11}{2} - 1\right| = \left|\frac{9}{2}\right| = \frac{9}{2}$$

72. Number of reflexive relations $= 2^{n^2 - n}$

73. Apply centroid formula

74. $$\sum_{r=0}^{n}\left({}^nC_r + \frac{2}{r+1}{}^nC_r\right) = ({}^nC_0 + {}^nC_1 + {}^nC_2 \ldots\ldots {}^nC_n) + 2\left[\frac{{}^nC_0}{1} + \frac{{}^nC_1}{2} + \frac{{}^nC_2}{3} \ldots\ldots \frac{{}^nC_n}{n+1}\right]$$

.

$$= (1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \ldots\ldots {}^nC_n x^n$$

$$= \int_0^1 (1+x)^n dx = \int_0^1 (C_0 + C_2 x + C_2 x^2 \ldots . C_n x^n) dx .$$

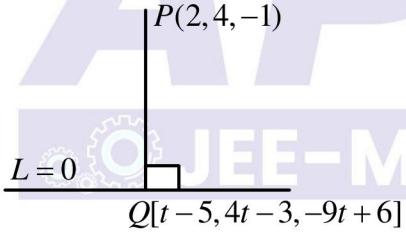
$$= \frac{2^{n+1} - 1}{n+1} = \frac{C_0}{1} + \frac{C_1}{2} + \ldots\ldots + \frac{C_n}{n+1} .$$

$$\sum_{r=0}^{n}\left(\frac{r+3}{r+1}\right){}^nC_r = 2^n + \frac{(2^{n+1} - 1)2}{n+1} = \frac{(n+1)2^n + 4 \cdot 2^n - 2}{n+1} .$$

$$= \frac{(n+5)2^n - 2}{n+1} = \frac{7 \cdot 2^{10} - 2}{10} = \frac{14 \cdot 2^9 = 2}{10}$$

$n = 9$.

75.



$\overrightarrow{PQ} \perp L = 0$

$a(a_1) + b(b)_2 + c(c_2) = 0$

Find 't' substitute in Q

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